

Natural lipid-lowering agents and their effects: an update

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Abstract Hyperlipidemia is one of the main contributing factors in the development of cardiovascular diseases. The hunt for effective antihyperlipidemic drugs is still on with an intention of finding potent agents with meager adverse effects. For the past several decades, medicinal plants have been evaluated for lipid-lowering effects apart from synthetic approaches to develop newer antihyperlipidemic agents. This article highlights the work carried out on potential antihyperlipidemic agents from plant sources from 2006 till 2012 that were archived in Google Scholar and Science Direct database. Seventy-nine articles were included in the review after manual content curation. The review summarizes the investigations on medicinal plants, compounds and herbal formulations using various hyperlipidemic animal models or in humans, the research findings and possibility of chemotaxonomic occurrence of secondary metabolites responsible for the antihyperlipidemic activities within certain plant families.

Keywords Antihyperlipidemic · Hypolipidemia · Medicinal plants · Natural products

Abbreviations

ACAT	Acyl-coenzyme A: cholesterol acyltransferase
AI	Atherogenic index
ASCVD	Atherosclerotic cardiovascular diseases
CRI	Coronary risk index

FBG	Fasting blood glucose
FFA	Free fatty acids
HDL-C	High-density lipoprotein cholesterol
HFD	High-fat diet
HMG-Co A	3-Hydroxy-3-methylglutaryl coenzyme A
h	Hours
I.P	Intra peritoneal
LCAT	Plasma lecithin cholesterol acyl transferase
LDL-C	Low-density lipoprotein cholesterol
LPL	Lipoprotein lipase
P.O	Per oral
STZ	Streptozotocin
TC	Total cholesterol
TG	Triglycerides

Introduction

Hyperlipidemia is defined as an increase in the blood lipids levels, which include cholesterol and triglycerides. It contributes significantly to the manifestation and development of atherosclerosis and coronary heart diseases. Cardiovascular diseases, including atherosclerosis, are among the most common causes of mortality and morbidity worldwide [1, 2]. Globally, each year approximately 12 million people die due to cardiovascular diseases. Factors such as diet high in saturated fats and cholesterol, age, family history, hypertension and life style are of great significance but high level of cholesterol, particularly LDL cholesterol is mainly responsible for the occurrence of coronary heart disease [3].

Persons with familial hypercholesterolemia commonly develop premature atherosclerosis and clinical atherosclerotic cardiovascular diseases (ASCVD) even in the absence of other risk factors [4]. Adverse risk factors induced by

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obesity and physical inactivity can aggregate to produce a multiplex risk factor for ASCVD and diabetes called the metabolic syndrome. This syndrome consists of atherogenic dyslipidemia (high TG and low HDL-C), high blood pressure, increased plasma glucose, a prothrombotic state and a proinflammatory state. In many countries, the prevalence of the metabolic syndrome ranges between 20 and 30 % of the adult population while in some populations, the prevalence can be even higher [5]. Hyperlipidemia is also known to aggravate several other pathological conditions such as hypothyroidism and chronic kidney dysfunction in combination with their respective underlying primary etiological factors of the disorders [6].

There are various classes of synthetic lipid-lowering agents employed in current management of hyperlipidemia, belonging to the statins, fibrates or bile acid sequestrants groups. Although they possess beneficial therapeutic effects, they are often associated with some serious side effects such as rhabdomyolysis, myopathy and an increased risk of gallstones. Thus, there is an exigent need for new lipid-lowering agents with high therapeutic value and minimum tolerable side effects [7]. This article attempts to summarize the research works for the past 7 years in the field of medicinal plants for the development of natural lipid-lowering agents.

Methods

The strategy adopted was to search for the relevant articles archived in the Google Scholar and Science Direct database using designated key words, namely ‘antihyperlipidemia’ and ‘hypolipidemia’. The database was searched for publications between the beginning of February 2006 until the end of 2012. Initial search of the database yielded 406 articles related to the key words. However, after a search within the results using secondary key words within the scope of this review, namely ‘medicinal plants’ and ‘natural products’, 79 articles were obtained. Book and reference work were excluded in this review. Since the publications encompass wide range of secondary metabolites, extracts and plant families, this review was arranged and described according to the plant parts such as leaves, fruit, bark, flowers and whole plant.

Findings

From the database searched, a total of 79 articles describing the antihyperlipidemic activity from the plant sources inclusive of compounds and complex formulations were found from 2006 till 2012. The contribution of different parts of medicinal plants to publications related to lipid-lowering potentials of medicinal plants/natural products from year 2006 till 2012 is summarized in Fig. 1.

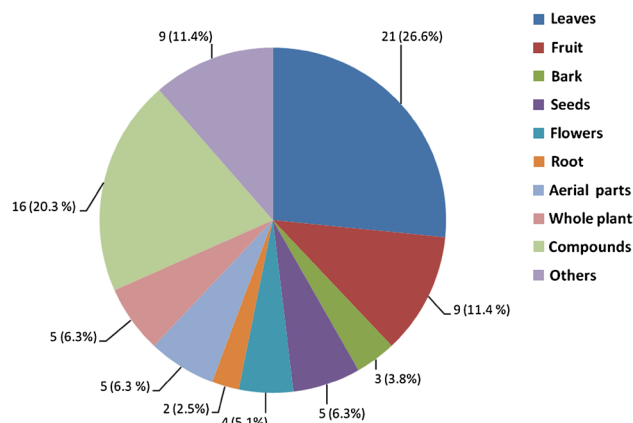


Fig. 1 Numbers and percentages (in bracket) of publications related to lipid-lowering potentials of medicinal plants/natural products from year 2006 till 2012 classified according to different plant parts. A total of 79 publications were found from the database

As depicted in Fig. 1, 21 articles comprising around 26.6 % of total publications assessed the leaves extracts alone. This observation could be related to the way the plants are used traditionally, whereby the leaves are the most preferred part. Leaves are usually consumed raw as salad in daily diet or in the form of decoction against various ailments. The finding suggests that the leaves of the medicinal plants may possess the secondary metabolites responsible for the antihyperlipidemic activity compared with other parts of the medicinal plants. The secondary metabolites occurrence seems to be probably higher in concentration in the leaves, but could also be found in other parts of the plant at lower concentration. This fact is substantiated not only from good number of publications from leaves but also publications from other metabolically active plant parts such as the fruit, aerial parts and whole plants.

The least number of publications was related to root and bark. This might be because of the difficulties in collecting these plant parts, in particular the lower yield of these parts as well as seasonal occurrence that could be limiting factors for their use in research. In the case of root and flower, probably it could be due to lack of secondary metabolites responsible for this pharmacological activity. The flower and seeds usually contain lesser chemical constituents, such as anthocyanidins, phenolic, coloring pigments, protein and lipids.

The number of publications employing pure compounds was found to be lesser compared with the use of plant parts or the whole plant. This could be the result of two factors; firstly, adversity in isolating adequate amount of compounds for pharmacological evaluation using laboratory animals or mechanistic studies. Secondly, the fact that a wide range of polar and non-polar chemical constituents are present in the crude extract or active fraction, which together act by multiple mechanisms such as enhancing the

lipid excretion, bile acid secretion, antioxidant activity and influencing the metabolism of lipoproteins, therefore capable of producing potent activity in comparison with single compound. But this cannot be certainly taken as an indicator that the pure compound possess weak pharmacological activity or do not contribute significantly to antihyperlipidemic activity. A well-known success story of a single compound is lovastatin, isolated from microbe *Aspergillus terreus* fungus. Lovastatin had potent HMG-Co A inhibiting activity and made significant contribution in discovery of statins [8].

Review of lipid-lowering studies according to different parts of the plants

In this review, the lipid-lowering studies will be discussed according to the plant parts used in those studies. Table 1 summarizes the medicinal plants having antihyperlipidemic activity while Table 2 summarizes the chemical compounds, formulations and nutraceuticals having antihyperlipidemic activity. Details such as extraction solvent, animal models used, study duration, the classification of whether the study involved antihyperlipidemic, mechanistic or toxicological investigations, the dose and route of administration as well as the limitations of the studies are summarized in the tables and will be described in greater details in the following sub-sections.

Leaves

High-fat diet-induced hyperlipidemic model

Antihyperlipidemic effects of the methanolic extract of *Vernonia amygdalina* leaves were examined in rats fed on high-cholesterol diet. Treatment with the methanolic extract of *V. amygdalina* caused a dose-dependent reduction in the plasma and post-mitochondrial fraction cholesterol levels by 20 and 23 % at 100 mg/kg and 23 and 29 % at 200 mg/kg, respectively. Furthermore, *V. amygdalina* at 200 mg/kg decreased the plasma and post-mitochondrial fraction LDL-C levels by 23 and 49 % and decreased plasma and post-mitochondrial fraction TG levels by 29 and 28 %, respectively. In contrast, *V. amygdalina* at 100 and 200 mg/kg caused a dose-dependent increase in plasma HDL-C levels by 41 and 59 %, respectively. There were significant decreases ($P < 0.05$) noticed in the lipid peroxidation levels of extract-treated rats. In addition, *V. amygdalina* augmented the cholesterol-induced decrease in post-mitochondrial fraction glutathione levels of the rats [9].

Ochani and D'Mello examined the antihyperlipidemic effects of *Hibiscus sabdariffa* leaves and calyces extracts in cholesterol-induced hyperlipidemic rats. The extracts

significantly reduced the levels of TC, LDL-C, VLDL-C and TG, but the level of HDL-C was increased. The treated animals had a significant decrease in the AI, a risk ratio of atherosclerosis [10]. Likewise, *Ocimum sanctum* L. leaves extract significantly reduced the high serum levels of TC, LDL-C and AI ($P < 0.001$) of high-cholesterol diet-fed rats. However, the levels of TG and HDL-C were not affected [11].

The aqueous extract of *Erythrina indica* leaves, at two dose levels of 200 and 300 mg/kg, administered for 30 days resulted in the reductions of TC, TG, LDL-C levels and a significant increase in HDL-C level, in high-fat diet-fed rats. The results were found to be comparable to that of standard drug, simvastatin at 5 mg/kg [12]. Effect of the aqueous leaves extracts of *Cyphostemma glaucophylla* on lipid profile was evaluated in albino rats fed with standard pellets diet. Significant ($P < 0.05$) lipid-lowering effect of the extract was found at the highest dose of 20.0 mg/kg. The percentage decreases in TC, TG, LDL-C and VLDL-C were 20.8, 65.8, 36.3 and 37.6, respectively, whereas the maximum percentage increase in HDL-C was 51.9 % [13].

Lakshmi et al. studied the antihyperlipidemic effects of *Bauhinia purpurea* in high-cholesterol diet-fed rats. The ethanolic extracts (300 mg/kg) of *B. purpurea* leaves and unripe pods showed significant reductions ($P < 0.05$) in TC, TG, LDL-C and VLDL-C levels and a significant increase in HDL-C level [14]. In 2012, Verma et al. [15] evaluated the antihyperlipidemic effect of an aqueous extract from *Pachyptera hymenaea* leaves administered P.O for 28 days in high-fat diet-induced hypercholesterolemic rats. It was found that the extract resulted in significant declines in TC, TG and LDL-C of 28.1, 27.9 and 44.0 % at 200 mg/kg and of 66.1, 57.6 and 60.0 % at 400 mg/kg. It was concluded that the extract possesses pronounced hypolipidemic and antihyperlipidemic effects, which were comparable to those of atorvastatin and these effects are due to the presence of organosulfur compounds, flavonoids and polyphenols in the extract [15].

Chemically-induced hyperlipidemic model

Total methanolic extract of *Salvadora oleoides* L. leaves was examined for its antihyperlipidemic activity in triton WR-1339-induced hyperlipidemic rats. Administration of the extract at a dose level of 100 mg/kg resulted in significant reductions ($P < 0.05$) in TC and TG levels as compared to normal rats and standard drug, fenofibrate [16]. Prasad et al. isolated five bioactive compounds from the leaves of *Calophyllum inophyllum* (calophyllic acid, isocalophyllic acid mixture, canophyllic acid, amentoflavone and shikimic acid) and evaluated their antidyslipidemic activity in triton-induced hyperlipidemia model at doses of 50, 100 and 150 mg/kg for 18 h. All compounds except the shikimic acid showed dose-dependent lipid-lowering activity.

Table 1 Medicinal plants having potential antihyperlipidemic activity

No.	Family name	Plant name	Part used	Extraction solvent	Animal models	Study duration	Scope of study	Dosage and route of administration	Limitations	References
1	Amaranthaceae	<i>Amaranthus caudatus</i>	Leaves	Methanol	Triton WR-1339-induced hyperlipidemic rats	48 hrs	Antihyperlipidemic activity	200, 300 and 400 mg/kg P.O	No description on antihyperlipidemic mechanism	[18]
		<i>Amaranthus spinosus</i>	Leaves	Methanol	Triton WR-1339-induced hyperlipidemic rats	48 hrs	Antihyperlipidemic activity	200, 300 and 400 mg/kg P.O	No description on antihyperlipidemic mechanism	[18]
		<i>Amaranthus viridis</i>	Leaves	Methanol	Triton WR-1339-induced hyperlipidemic rats	48 hrs	Antihyperlipidemic activity	200, 300 and 400 mg/kg P.O	No description on antihyperlipidemic mechanism	[18]
		<i>Amaranthus caudatus</i> L.	Aerial parts	Ethanol	HFD-induced hyperlipidemic rabbits	75 days	Antihyperlipidemic and antiatherogenic activities	150 mg/kg P.O	Positive control was not employed in the study	[40]
		<i>Amaranthus viridis</i>	Whole plant	Methanol	Alloxan-induced diabetic rats	15 days	Antidiabetic, antihyperlipidemic and antioxidant activities	200 and 400 mg/kg P.O	Diabetic model used to study antihyperlipidemia	[45]
2	Apiaceae	<i>Coriandrum sativum</i>	Seed	Methanol	Cholesterol-induced hyperlipidemic rabbits	120 days	Antioxidant and lipid-lowering activities	500 mg/kg P.O	Positive control was not employed in this study	[59]
		<i>Peucedanum pascuaticum</i>	Aerial parts	Ethanol	Cholesterol-enriched diet-induced hypercholesteremic rats	56 days	Antihyperlipidemic activity	125, 250 and 500 mg/kg P.O	No description on antihyperlipidemic mechanism	[38]
		<i>Peucedanum pascuaticum</i>	Aerial parts	96 % ethanol	STZ-induced diabetic rats	30 days	Antihyperlipidemic activity	125, 250 and 500 mg/kg P.O	No description on antihyperlipidemic mechanism	[37]
3	Apocynaceae	<i>Allamanda villosa</i>	Flower	Aqueous	Triton WR-1339 and fructose-rich-induced hyperlipidemic rats	18 hrs and 30 days respectively	Antihyperlipidemic, antioxidant and antidiabetic activities	100 and 50 mg/kg P.O	Short duration for acute study	[32]
4	Araliaceae	<i>Panax notoginseng</i>	Root	NM	HFD-induced hyperlipidemic rats	4 weeks	Antihyperlipidemic activity	0.25, 0.5 and 1 % (w/w)	Equal consumption of Sanchi diet supplement may not be ensured	[55]
5	Asteraceae	<i>Vernonia amygdalina</i>	Leaves	Methanol	High-cholesterol diet-induced hyperlipidemic rats	9 weeks	Antihyperlipidemic activity	100 and 200 mg/kg P.O	–	[9]
		<i>Cynara scolymus</i>	Leaves	Aqueous	STZ-induced diabetic rats	21 days	Hypolipidemic and hypoglycemic activities	200 and 400 mg/kg P.O	Positive control was not employed in this study	[25]
		Artichoke Leaves	Artichoke Leaves	NM	Lipogenic diet-induced hyperlipidemic rats	60 days	Antihyperlipidemic activity	10 % artichoke mixed with standard diet	Positive control was not employed in this study	[34]

Table 1 continued

No.	Family name	Plant name	Part used	Extraction solvent	Animal models	Study duration	Scope of study	Dosage and route of administration	Limitations	References
		<i>Melanthera scandens</i>	Leaves	Ethanol	Alloxan-induced diabetic rats.	7 hrs in acute study and 14 days in chronic study	Antidiabetic and hypolipidemic activities	37, 74 and 111 mg/kg P.O	No description on antihyperlipidemic mechanism	[23]
		<i>Eclipta prostrata</i>	Whole plant	Ethanol	HFD-induced hyperlipidemic rats	30 days	Antihyperlipidemic and antioxidant activities	50, 100, 150 and 200 mg/kg P.O	No description on antihyperlipidemic mechanism	[47]
6	Bignoniaceae	<i>Pachyptera hymenaea</i>	Leaves	Aqueous	HFD-induced hypercholesterolemic rats	28 days	Hypolipidemic and antihyperlipidemic activities	200 and 400 mg/kg P.O	No description on antihyperlipidemic mechanism	[15]
7	Calophyllaceae	<i>Calophyllum inophyllum</i>	Leaves	Ethanol	Triton WR 1339-induced hypercholesterolemic rats	18 hrs	Antidyslipidemic and antioxidant activities	50, 100, 150 mg/kg P.O.	Short duration of the study and no description on mechanism	[17]
8	Celastraceae	<i>Celastrus paniculatus</i>	Seed	Methanol	HFD-induced hypercholesterolemic rats	6 weeks	Antihyperlipidemic activity	25, 45, 65 and 75 mg/kg P.O	No description on antihyperlipidemic mechanism	[58]
9	Combretaceae	<i>Terminalia chebula</i> and <i>gaumutra</i> <i>Terminalia arjuna</i>	Fruit Bark	Aqueous Petroleum ether, then fractions (methanol, diethyl ether, ethyl acetate)	HFD-induced hypercholesterolemic rats Poloxamer 407-induced hypercholesterolemic mice	30 days 18 hrs	Antihyperlipidemic activity Antihyperlipidemic and antioxidant activities	300, 500 and 700 mg/kg P.O 175 and 350 mg/kg P.O	No description on antihyperlipidemic mechanism Short duration of the study and no description on mechanism	[49] [30]
10	Cucurbitaceae	<i>Coccinia cordifolia</i>	Leaves	Ethanol	STZ-induced diabetic rats	24 hrs	Antidiabetic and hypolipidemic activities	150 mg/kg I. P	The investigation necessitate the dose response study, short duration of the study and no description on mechanism of action	[25]
		<i>Lagenaria siceraria</i>	Fruit	Juice	Atherogenic diet-induced hypercholesterolemic rats	4 weeks	Antihyperlipidemic activity	200 mg/kg P.O	Inadequate number of animals were used, mechanism and there is no dose response, only one dose was used	[54]

Table 1 continued

No.	Family name	Plant name	Part used	Extraction solvent	Animal models	Study duration	Scope of study	Dosage and route of administration	Limitations	References
			Fruit	Methanol	HFD-induced hypercholesterolemic rats	30 days	Antihyperlipidemic activity	100, 200 and 300 mg/kg P.O	–	[50]
11	Ericaceae	<i>Erica multiflora</i> L.	Flower	Aqueous	Triton WR-1339-induced hypercholesterolemic rats	24 hrs	Antihyperlipidemic activity	250 mg/100 g P.O	The investigation necessitate the dose response study, short duration of the study and no description on mechanism of action	[35]
12	Fabaceae	<i>Erythrina indica</i>	Leaves	Aqueous	Cholesterol-enriched diet	8 weeks	Antihyperlipidemic activity	200 and 300 mg/kg P.O	No description on mechanism of action	[12]
		<i>Bauhinia variegata</i>	Leaves	Methanol	Triton WR-1339-induced hypercholesterolemic rats	7 days	Antihyperlipidemic activity	65 mg/kg P.O	There is no dose response, only one dose was used	[19]
		<i>Cassia auriculata</i> L.	Flower	Ethanol	Triton WR-1339-induced hypercholesterolemic rats	14 days	Antihyperlipidemic and antioxidant activities	150, 300 and 450 mg/kg P.O	–	[33]
		<i>Glycine tomentella</i>	Root	Ethanol	HFD Tilapia fish (<i>Oreochromis mossambicus</i>)	12 weeks	Hypolipidemic activity	1 % (w/w) extract	–	[56]
		<i>Arachis hypogaea</i>	Peanut skin	Water	HFD (western diet)-induced hyperlipidemic rats	10 weeks	Hypolipidemic activity	150 and 300 mg/kg 5 times a week for 10 weeks P.O	Positive control was not employed in this study	[71]
13	Lamiaceae	<i>Ocimum sanctum</i> L.	Leaves	Hydrodistillation to obtain essential oil from leaves	HFD-induced hypercholesterolemic rats	7 weeks	Antihyperlipidemic and antioxidant activities	80 µl/kg P.O	There is no dose response, only one dose was used	[11]
		<i>Orthosiphon stamineus</i>	Bark	Petroleum ether	HFD-induced hypercholesterolemic rats.	10 days	Antihyperlipidemic activity	500 and 750 mg/kg P.O	No description on mechanism of action	[28]
		<i>Marrubium vulgare</i>	Aerial parts	Methanol	STZ-induced diabetic rats	28 days	Antidiabetic and antidiabetic activities	500 and 750 mg/kg P.O	–	[39]
		<i>Ajuga reptans</i>	Whole plant	Aqueous	STZ-induced diabetic rats.	6 hrs for single dose and 21 days for sub-chronic study	Hypolipidemic activity	10 mg/kg P.O	There is no dose response, only one dose was used and no description on mechanism of action	[46]

Table 1 continued

No.	Family name	Plant name	Part used	Extraction solvent	Animal models	Study duration	Scope of study	Dosage and route of administration	Limitations	References
14	Lauraceae	<i>Cinnamomum zeylanicum</i>	Bark	Aqueous and methanol	HFD-induced hypercholesterolemic rabbits	180 days	Antihyperlipidemic activity	250, 500 and 750 mg/kg P.O	No description on mechanism of action	[29]
15	Leguminosae	<i>Bathinia purpurea</i>	Unripe pods and leaves	Ethanol	HFD-induced hypercholesterolemic rats	30 days	Antihyperlipidemic activity	300 mg/kg P.O	No description on mechanism of action	[14]
16	Malvaceae	<i>Abutilon indicum</i>	Leaves	Ethanol and aqueous	Triton WR-1339 and HFD-induced hyperlipidemic rats	24 hrs and 7 days	Antihyperlipidemic activity	400 mg/kg P.O	There is no dose response, only one dose was used and no description on mechanism of action	[20]
		<i>Hibiscus sabdariva</i>	Leaves	Ethanol	HFD-induced hypercholesterolemic rats	30 days	Antihyperlipidemic and antioxidant activities	500 mg/kg P.O	There is no dose response, only one dose was used and no description on mechanism of action	[10]
		<i>Helicteres isora</i>	Fruit	Ethanol	STZ-induced diabetic rats	45 days	Antidiabetic and antihyperlipidemic activities	300 mg/kg P.O	There is no dose response, only one dose was used and no description on mechanism of action	[53]
17	Myrtaceae	<i>Callistemon lanceolatus</i>	Leaves	Ethyl acetate fraction from methanol	Alloxan-induced diabetic rats	21 days	Antihyperglycemic and antihyperlipidemic activities	200 and 400 mg/kg P.O	No description on mechanism of action	[22]
		<i>Psidium guajava puree</i>	Fruit	Aqueous	HFD-induced hypercholesterolemic rats.	6 weeks	Antibesity activity	500, 1,000 and 2,000 mg/kg orally with water	Doses used were rather high	[48]
18	Nelumbo-naceae	<i>Nelumbo nucifera</i>	Flower petals	Dried flower petals powder	Young, middle-aged and aged rats	4 weeks	Changes in plasma lipids and glucose	1,000 mg/kg bi dose/day P.O	There is no induction of hyperlipidemia	[31]
19	Oxalidaceae	<i>Averrhoa bilimbi</i>	Fruit	Aqueous	Triton x-100 and HFD-induced hypercholesterolemic rats	24 hrs and 60 days	Antihyperlipidemic activity	125 and 250 mg/kg P.O	No description on mechanism of action	[51]

Table 1 continued

No.	Family name	Plant name	Part used	Extraction solvent	Animal models	Study duration	Scope of study	Dosage and route of administration	Limitations	References
20	Pedaliaceae	<i>Sesamum indicum</i>	Fat free seeds powder	–	HFD-induced hypercholesterolemic rats	4 weeks	Antihyperlipidemic and antioxidant activities	5 and 10 % s. indicum seed powder mixed with diet	–	[57]
21	Pleurotaceae	<i>Pleurotus citrinopileatus</i>	Whole mushroom	Aqueous (ethyl acetate and methanol)	HFD-induced hypercholesterolemic rats	42 days	Antihyperlipidemic and antioxidant activities	80, 200, 500 and 2,000 mg/kg P.O	Positive control has not been employed in the study. Investigation of mechanism of action is lacking	[41]
22	Poaceae	<i>Zea mays</i> L. (Corn silk)	Flavonoid	Ethanol	HFD-induced hypercholesterolemic rats	20 days	Antihyperlipidemic and antioxidant activities	200, 400 and 800 mg/kg P.O	Investigation of mechanism of action is lacking	[73]
23	Polyporaceae	<i>Coriolus versicolor</i>	Whole mushroom	Aqueous	Poloxamer-407 and HFD-induced hypercholesterolemic rats	3 and 14 days	Antihyperlipidemic activity	500 and 1,000 mg/kg P.O	Investigation of mechanism of action is lacking	[44]
24	<i>Pseudanabaena-eae</i>	<i>Spirulina platensis</i>	Cyanobacterium	Ethanol	Triton X-100-induced hypercholesterolemic rats	7 days	Antihyperlipidemic activity	500 mg/kg P.O	Investigation of mechanism of action is lacking	[46]
25	Ranunculaceae	Kalonji (<i>Nigella sativa</i>)	Seed	Seed	Patients with high lipid profile	4 weeks	Antihyperlipidemic activity	Two tea-spoons of Kalonji after breakfast per day P.O	Investigation of mechanism of action is lacking	[68]
26	Rosaceae	<i>Pyrus bioessieriana</i> Buhse	Leaves	Methanol	Alloxan-induced rat model of hyperglycemia	72 hrs	Antihyperglycemia and antihyperlipidemic activities	500 and 1,000 mg/kg P.O	Positive control was not employed and no description on mechanism of action	[24]
27	Rubiaceae	<i>Anthocephalus indicus</i>	Fruit	Ethanol	Triton WR-1339 and HFD-induced hypercholesterolemic rats	30 days	Antihyperlipidemic activity	200 and 500 mg/kg P.O	No description on mechanism of action	[52]
		<i>Randia dumetorum</i>	Fruit	Methanol	Triton WR-11339-induced hypercholesterolemic rats	2 days	Antihyperlipidemic activity	400 mg/kg P.O	No description on mechanism of action	[36]
		<i>Paederia foetida</i>	Aerial part	Methanol	Triton WR-1339	2 days	Antihyperlipidemic activity	400 mg/kg P.O	No description on mechanism of action	[36]
28	Rutaceae	<i>Aegle marmelos</i>	Seed	Aqueous	Normal, sub and mild diabetic rats	14 days	Antihyperlipidemic activity	100, 250 and 500 mg/kg P.O	No description on mechanism of action	[60]

Table 1 continued

No.	Family name	Plant name	Part used	Extraction solvent	Animal models	Study duration	Scope of study	Dosage and route of administration	Limitations	References
29	Sapindaceae	<i>Sapindus emarginatus</i>	Fruit	Methanol	Triton WR-1339	7 days	Antihyperlipidemic activity	100 and 200 mg/kg P.O	No description on antihyperlipidemic mechanism	[44]
30	Symplacaceae	<i>Symplocos cochinchinensis</i>	Leaves	Hexane, ethyl acetate, and methanol	Triton WR-1339 and HFD-induced hyperlipidemic rats	18 hrs and 28 days	Antihyperlipidemic activity	250 and 500 mg/kg P.O	No description on antihyperlipidemic mechanism and short study duration	[2]
31	Salvadoraceae	<i>Salvadora oleoides</i>	Leaves	Methanol	Triton WR-1339	7 days	Antihyperlipidemic activity	100 mg/kg P.O	No description on antihyperlipidemic mechanism	[16]
32	Verbenaceae	<i>Clerodendrum capitatum</i>	Leaves	Aqueous	Normal healthy rats	14 days	Antihyperlipidemic activity	100, 400 and 800 mg/kg P.O	No description on mechanism of action	[27]
33	Vitaceae	<i>Cyphostemma glaucophylla</i>	Leaves	Aqueous	Normal healthy rats	14 days	Antihyperlipidemic activity	5, 10, 15 and 20 mg/kg P.O	–	[15]

HFD high-fat diet, hrs hours, P.O per oral, STZ streptozotocin, NM not mentioned

At the same time, calophyllic acid, isocalophyllic acid mixture and canophyllic acid also exhibited good antioxidant activity [17].

Antihyperlipidemic activity of *Amaranthus* leaves extracts was examined using normal and triton WR-1339-induced rats at doses of 200, 300 and 400 mg/kg. The groups treated with the methanolic extracts of *A. caudatus*, *A. spinosus* and *A. viridis* showed significant decrease ($P < 0.01$) in the level of TC at 400 mg/kg after 24 and 48 h. In addition, *A. viridis* also showed significant decrease ($P < 0.05$) in the TC level at 300 mg/kg. The level of TG was significantly decreased ($P < 0.01$) by methanol extracts of *A. caudatus* and *A. viridis* at 400 mg/kg. Likewise, methanol extract of *A. viridis* at 300 mg/kg exhibited a significant reduction in TG ($P < 0.05$) after 24 and 48 h [18]. On the other hand, methanol extracts of *A. caudatus* and *A. spinosus* showed a significant increase in the level of HDL-C at 400 mg/kg ($P < 0.05$). Similar significant increase ($P < 0.05$) in HDL-C was also seen in methanolic extract of *A. viridis* at 300 mg/kg after 24 and 48 h. The level of LDL-C significantly decreased with the methanol extracts of *A. viridis* at 400 mg/kg ($P < 0.01$) and at 300 mg/kg ($P < 0.05$) after 24 and 48 h, respectively [18]. Girija and Lakshman concluded that *A. viridis* showed highly significant antihyperlipidemic activity compared with *A. caudatus* and *A. spinosus* in triton WR-1339-induced hyperlipidemic rats [18].

Combination of high-fat- and chemically-induced hyperlipidemic model

The evaluation of hypolipidemic activity-guided fractionations of methanolic extract of *Bauhinia variegata* L. leaves was performed in triton WR-1339-induced hyperlipidemic rats. By applying increasing polarity of solvents in the separation technique, butanol fraction yielded four sub-fractions (A–D). At dose of 65 mg/kg (P.O), the butanol sub-fraction (D) was effective as antihyperlipidemic agent, which resulted in significant reductions ($P < 0.05$) in TC, TG, LDL and VLDL-C levels and also increases the HDL-C level [19].

Giri et al. reported the antihyperlipidemic effects of *Abutilon indicum* in triton WR-1339 and diet-induced hyperlipidemic Wistar albino rats. The ethanolic and water extracts of *A. indicum* leaves at 400 mg/kg dose reduced the increase in TC and TG levels in triton WR-1339 group. Likewise, it inhibited the increase in TC and TG but increased HDL-C in high-fat diet-induced rats [20]. The hypolipidemic activity of *Symplocos cochinchinensis* leaves was studied in high-fat diet and triton WR-1339-induced hyperlipidemic rats. The hexane extract (250 and 500 mg/kg) exerted significant ($P < 0.01$) lipid-lowering effect compared with ethyl acetate and methanolic extracts.

After treatment with *S. cochinchinensis* for 28 days, the plasma levels of TC ($P < 0.01$), TG ($P < 0.01$) and LDL-C ($P < 0.01$) were significantly reduced, while the level of HDL-C ($P < 0.01$) was significantly increased [21].

Other hyperlipidemic model

Ethyl acetate fraction of *Callistemon lanceolatus* ethanolic leaves extract was administrated in alloxan-induced diabetic rats (200 and 400 mg/kg) for 21 days. It significantly decreased the levels of TC and TG ($P < 0.05$) and enhanced the HDL-C level [22]. Akpan et al. examined the hypolipidemic activities of leaves ethanolic extract and fractions of *Melanthera scandens* in alloxan-induced diabetic rats. *M. scandens* leave extract and fractions (37–111 mg/kg) were administered for 2 weeks. Ethanolic extract and fractions exhibited significant reductions in lipid profiles, TC, TG, LDL and VLDL-C while increased the HDL-C level of the diabetic rats [23]. Shahaboddin et al. evaluated the efficacy of methanolic extract of leaves of *Pyrus brossieriana* Buhse in alloxan-induced rat model of hyperglycemia. The leaf extracts at doses of 500 and 1,000 mg/kg were found to be capable of reducing TG and TC levels and increasing the serum antioxidant levels. These activities of extracts were attributed mainly to the presence of hydroquinone glycoside, arbutin [24].

The effects of petroleum ether, ethyl acetate and chloroform fractions obtained from ethanolic extract of the leaves of *Coccinia cordifolia* L. at 150 mg/kg on lipid profiles were studied in normal and streptozotocin (STZ)-induced diabetic rats. Ethyl acetate fraction was found to be the most effective, which reduced TC level by 31.0 and 36.7 % and the level of TG by 43.8 and 42.0 % in normal and STZ-diabetic rats, respectively [25]. Heidarian and Soofiniya assessed the effect of aqueous extract of *Cynara scolymus* leaves in STZ-induced diabetic rats at dose of 200 and 400 mg/kg administered orally for 21 days. The extract ameliorates the level of plasma malondialdehyde and increased plasma antioxidant capacity of treated animals. The results also reflect the beneficial effects on reducing serum TC, TG, LDL-C, VLDL-C and glucose levels [26].

Study by Adeneye et al. reported the hypolipidemic effects of aqueous extract of *Clerodendrum capitatum* leaves at oral doses of 100, 400 and 800 mg/kg for 14 days [27]. The finding of the study revealed that *C. capitatum* extract caused significant dose-dependent hypolipidemic effects ($P < 0.05$) but had no effect in the weight gain of the treated rats [27].

Bark

Umbare et al. investigated the lipid-lowering effect of hydro-alcoholic extract of *Orthosiphon stamineus* Benth

in cholesterol diet-induced hyperlipidemic rats. The extract significantly reduced the levels of TC ($P < 0.01$) and TG ($P < 0.05$) in the serum and TC ($P < 0.05$) and TG ($P < 0.01$) of the liver tissue at doses of 500 and 750 mg/kg [28]. Lipid-lowering effects of *Cinnamomum zeylanicum* (cinnamon) were assessed in hyperlipidemic albino rabbits using high-fat diet. The results suggested that *C. zeylanicum* bark powder at the doses of 500 and 750 mg/kg and methanol extract equivalent to 750 mg/kg powder produced percentage reductions in total lipids by 45, 49 and 64 %, TG by 38, 53 and 60 %, TC by 53, 64, 69 % and LDL-C by 50, 59 and 62 %, respectively. However, at these dosages, HDL level was increased by 42, 48 and 53 %, respectively. On the other hand, it was noticed that at low dose (250 mg/kg), *C. zeylanicum* water extract has no effect on lipid profile [29].

Antihyperlipidemic activity of diethyl ether, ethyl acetate and ethanol fractions of *Terminalia arjuna* bark was investigated in poloxamer 407-induced hyperlipidemic rats at doses of 175 and 350 mg/kg for 24 h. Among these fractions, ethanolic fraction was found to have significant effect on TC and TG levels at both doses at 24 h ($P < 0.01$), but not on LDL-C and VLDL-C levels. HDL-C level was significantly increased ($P < 0.01$) at 24 h at the higher dose [30].

Flowers

Bhuvana et al. examined the antihyperlipidemic effects of *Nelumbo nucifera* flowers at 1,000 mg/kg twice daily for 4 weeks in albino rats of Wistar strains of different ages. Administration of *N. nucifera* prevented the age-associated changes in TC, TG, free fatty acids (FFA), phospholipids, LDL-C, VLDL-C and glucose. HDL-C level was increased significantly in young, middle-aged and aged rats after treatment with *N. nucifera*. These findings demonstrated that *N. nucifera* normalized the age-associated alteration levels of lipids in middle-aged and aged rats and thus can decrease the risk factors for cardiovascular diseases during aging process [31].

Fractions of aqueous extract of *Allamanda violacea* flower were examined for their antihyperlipidemic activity in triton WR-1339-induced hyperlipidemic rats. Among the fractions, ether and chloroform fractions caused marked decrease in the levels of TC (24 %) and TG (23 %). Similarly, in rats fed with high-fat diet, ether and chloroform fractions lowered TC (26 and 25 %) and TG (18 and 19 %), respectively [32]. Vijayaraj et al. investigated the effect of ethanolic extract of *Cassia auriculata* flowers in triton WR-1339-induced hyperlipidemic rats for 14 days. Significant reductions were found in TC (34.11 %), TG (44.7 %), LDL-C (60 %) and VLDL-C (44.7 %) levels ($P < 0.001$), while a significant increase was observed in HDL-C (85 %) after 14 days treatment ($P < 0.001$) compared with the

hyperlipidemic control group. AI was also significantly decreased (76.2 %) in hyperlipidemic rats administered with 450 mg/kg of extract ($P < 0.001$). Maximum antihyperlipidemic effect ($P < 0.001$) was observed at dose 450 mg/kg, which had better antihyperlipidemic effect than lovastatin [33].

Artichoke (*C. scolymus* L.) is reported to possess lipid-lowering activity when tested in rats fed with lipogenic diet. Significant decreases in liver TG ($P < 0.05$ and $P < 0.05$), the ratio of TC to HDL-C ($P < 0.05$ and $P < 0.01$) and plasma TC ($P < 0.05$ and $P < 0.01$) were observed after administration of artichoke 10 % in control supplemented group and 10 % artichoke hyperlipidemic-treated group, respectively. However, TG level was significantly decreased only with 10 % artichoke hyperlipidemic-treated group compared with normal control group ($P < 0.01$) [34].

Triton WR-1339-induced hyperlipidemic rats model (200 mg/kg i.p) was used to study the hypocholesterolemic and hypotriglyceridemic effect of aqueous extract from *Erica multiflora* L flower. After 7 h treatment, significant decreases ($P < 0.001$) were detected in TC, TG and LDL-C by 47, 95 and 67 %, respectively, but no significant change was observed in HDL-C level [36]. On the other hand, the percentages of reduction after 24 h of TC, TG and LDL-C were 68.5, 91 and 80.5 %, respectively, while no significant increase in HDL-C was observed [35]. In fenofibrate treated rats, the hypolipidemic effect of fenofibrate was limited to TG and LDL-C (92 and 41 %), respectively at 7 h. While after 24 h, only TG concentration was significantly lowered by 82 %. It was concluded that the aqueous extract may contain polyphenolic compounds that are capable to lower plasma lipid concentrations [35].

Aerial parts

Methanolic extract of the aerial parts of *Paederia foetida* has been studied in triton WR-1339-induced hyperlipidemia male rats at dose of 400 mg/kg for 2 days. It lowered TC from 157.47 mg/dL at 18 h to 133.15 mg/dL after 24 h. However, it showed no significant decrease in TG levels at both time points [36].

STZ-induced diabetic rats treated with hydroalcoholic extract of the aerial parts of *Peucedanum pastinacifolium* for 30 days had significant reductions in serum TC, TG and LDL-C levels, whereas the HDL-C level was significantly increased compared with control diabetic rats. The AI in groups treated with hydroalcoholic extract was significantly decreased compared with the diabetic control rats [37]. On the other hand, the ethanolic extract of aerial parts of *P. pastinacifolium* was studied in the cholesterol-enriched diet-fed rats by for 56 days. *P. pastinacifolium* extract significantly reduced TC, LDL-C, TG levels and AI in hypercholesterolemic rats ($P < 0.05$); however, the HDL-C level

Table 2 Compounds, formulations and nutraceuticals with potential antihyperlipidemic activity

No.	Name	Classification (drug, compound, food)	Animal models	Study duration	Scope of study	Dosage and route of administration	Limitations	References
1	Soy drink enriched with plant sterols	Food	Moderately hypercholesterolemic subjects (men and women)	8 weeks	Antihyperlipidemic activity	200 ml of soy drink either enriched with 2.6 g plant sterol esters (1.6 g/d free plant sterol equivalents) or without plant sterols	–	[64]
2	Apricot and pumpkin	Food	HFD-induced hyperlipidemic rats	28 days	Antihyperlipidemic activity	1,000 mg/kg P.O	Positive control not employed in this study and no description on mechanism of action	[65]
3	A poly herbal drug (Geriforte)	Drug	HFD-induced hyperlipidemic rats	8 days	Antihyperlipidemic activity	Geriforte (450 mg/kg and Ayurslim (250 mg/kg) P.O	No description on mechanism of action	[62]
4	Allicin	Compound	HFD-induced hyperlipidemic mice	12 weeks	Antihyperlipidemic activity	5, 10 and 20 mg/kg P.O	No positive control was employed in the study and no description on mechanism of action	[66]
5	Berberine	Drug	Obese human subjects	12 weeks	Antihyperlipidemic activity	500 mg berberine 3 times/day P.O	–	[79]
6	Protocatechuic acid	Compound	Fructose-induced hyperlipidemic rats	21 days	Antihyperlipidemic activity	25 and 50 mg/kg, along with the drinking water	No positive control was employed in this study and no description on mechanism of action	[78]
7	High-sulfate content polysaccharide (ulvan)	Compound	High-cholesterol-rich diet-fed Kunming mice	21 days	Antihyperlipidemic activity	250 mg/kg P.O	Investigation of mechanism of action of candidate under evaluation is lacking	[67]
8	Polysaccharides from <i>Apostichopus japonicus</i>	Compound	High-fat diet-induced lipidemia on male wistar rats,	28 days	Antioxidant and antihyperlipidemic activities	200, 400 and 800 mg/kg P.O	Investigation of mechanism of action of candidate under evaluation is lacking	[68]
9	Chlorogenic acid and tetrahydrocurcumin alone and in combination	Compounds	STZ-nicotinamide-induced type 2 diabetic rats	45 days	Antihyperlipidemic activity	5 and 80 mg/kg P.O	Positive control not employed in this study	[76]
10	Hydroxyethyl methylcellulose	Food	HFD mice	6 weeks	Antihyperlipidemic activity	Hydroxyethyl methylcellulose (HEMC) with different viscosities	Positive control not employed in this study	[70]

Table 2 continued

No.	Name	Classification (drug, compound, food)	Animal models	Study duration	Scope of study	Dosage and route of administration	Limitations	References
11	Gallic acid isolated from fruit rind of <i>Terminalia bellerica</i>	Compound	STZ-treated diabetic rats	28 days	Antihyperlipidemic activity	5, 10 and 20 mg/kg P.O	–	[77]
12	Calophylllic acid, isocalophylllic acid mixture, canophyllic acid, amentoflavone and shikimic acid	Compounds	Triton WR 1339-induced hyperlipidemic rats	18 hrs	Antidyslipidemic and antioxidant activity	50, 100, 150 mg/kg P.O	Short duration of the study and investigation of mechanism of action of candidate under evaluation is lacking	[17]
13	Fatclean	Formula	HFD-induced hyperlipidemic rats	6 weeks	Hypolipidemic and body fat-lowering effects	5 % w/w supplement diet	Positive control not employed in this study	[63]
14	Aegeline and [N-(2-hydroxy-2- <i>p</i> -tolylethyl)-amides and N-(2-oxo-2- <i>p</i> -tolylethyl)-amide derivatives]	Compounds	Triton WR 1339-induced hyperlipidemic rats	18 hrs	Antihyperlipidemic activity	100 mg/kg P.O	Short duration of the study	[75]
15	Leucodelphinidin derivative	Compound	HFD-induced hyperlipidemic rats	90 days	Antihyperlipidemic activity	100 mg/kg P.O	Positive control not employed in this study and investigation of mechanism of action of candidate under evaluation is lacking	[72]
16	Acidic polysaccharide isolated from Korean red ginseng	Compound	Triton WR-1339 or corn oil intra-venously injected	3 days	Hypocholesteraeamic and antioxidant activities	100, 300, 500 and 1,000 mg/kg	Investigation of mechanism of action of candidate under evaluation is lacking	[69]
17	Flavonoid from <i>Morus alba</i> (mulberry) leaves	Total flavonoid	Triton WR-1339-induced hypercholesterolemic mice	24 hrs	Hypolipidemic activity	7.5, 15 and 30 mg/kg P.O	No description on mechanism of action and short duration of the study	[74]

HFD high-fat diet, hrs hours, P.O per oral, STZ streptozotocin

was not affected [38]. Both hydroalcoholic extract and ethanolic extracts of the plant showed good antihyperlipidemic activity, and coumarins and furanocoumarins were thought to be responsible for the activity [37, 38].

Elberry et al. studied the antihyperlipidemic effects of the methanolic extract of the aerial part of *Marrubium vulgare* in STZ-induced diabetic rats, at a daily single oral dose of 500 mg/kg for 28 days. *M. vulgare* significantly reduced plasma TC, TG and LDL-C, while the cardio-protective lipid, HDL-C level was increased [39]. Hydroalcoholic extract of the aerial parts of *Amaranthus caudatus* L. was tested for its antihyperlipidemic activity on high-cholesterol diet rabbit model. A dose of 150 mg/kg for 30 days significantly reduced TC, LDL-C, malondialdehyde, C-reactive protein while apo lipoprotein A and HDL-C were significantly increased [40].

Whole plant

Hu et al. investigated the antihyperlipidemic effects of the four different forms of mushroom *Pleurotus citrinopileatus*: the powdered dry fruiting body, hot-water extract, and two kinds of elutes obtained from ethyl acetate and methanol extracts in high-fat diet-induced hyperlipidemic rats. Serum TG and TC levels were significantly lowered in groups supplemented with the highest dose of ethyl acetate and methanol extracts (500 mg/kg). HDL-C levels in these groups were also significantly higher than those in the control group [41]. The rats that were fed with ethyl acetate extract had the highest serum glutathione peroxidase and superoxide dismutase activities, and those fed with the methanol or ethyl acetate extracts had the highest DPPH free radical-scavenging activities and ferric-reducing abilities. Phytochemical analysis showed that the major constituents of methanol and ethyl acetate extracts were ergosterol and nicotinic acid, respectively. The antioxidant activities and antihyperlipidemic effects of methanol and ethyl acetate extracts were similar in potencies [41].

Sapinduse marginatus was examined for its antihyperlipidemic activity as it contains saponins in high quantities, and saponins were reported to exhibit antihyperlipidemic activity. The extract was tested in triton WR-1339-induced hyperlipidemic rats at doses of 100 and 200 mg/kg administered orally. The extract significantly decreased the levels of serum TC, phospholipids, TG, LDL-C and VLDL-C, while significantly increased the level of serum HDL-C. Methanol extracts at 200 mg/kg decreased serum level of TC by 69.7 %. On the other hand, the aqueous extract increased the serum HDL-C level by 24.1 % and found to decrease LDL-C level by 30.3 %. The activity of this plant was hypothesized due to the presence of saponins and flavonoids [42].

The ethanolic extract of *Spirulina platensis* was administered at a dose of 500 mg/kg, orally to triton WR-1337-induced hyperlipidemic rats. The extract significantly decreased the levels of serum TC, phospholipids, TG, LDL-C and VLDL-C and significantly increased the level of serum HDL-C [43]. Hor et al. evaluated the antihyperlipidemic effect of *Coriolus versicolor* extracts in poloxamer 407-induced hypercholesterolemic and high-cholesterol-fed rats. *C. versicolor* extract significantly reduced serum TC ($P < 0.01$) and TG ($P < 0.05$) levels. It also reduced the coronary risk index (CRI) in these rats. In case of the diet-induced hypercholesterolemic rat model, the serum TC and LDL-C levels were significantly reduced in a dose-dependent manner in rats given *C. versicolor* extract orally compared with the rats fed a high-cholesterol diet for 14 days. The administration of *C. versicolor* extract also significantly increased HDL-C level and reduced the CRI ratio [44].

Methanolic extract of whole plant of *Amaranthus viridis* was studied for its antihyperlipidemic activity in alloxan-induced diabetic rats. *A. viridis* methanolic extract (200 and 400 mg/kg) significantly reduced the levels of TG, TC, LDL-C and VLDL-C [45]. El-Hilalya et al. evaluated the lipid-lowering effect of an aqueous extract of whole plant of *Ajuga iva* L. (Lamiaceae) in normal and STZ-diabetic rats. This study was divided into acute (single dose for 6 h) and sub-chronic (daily dose for 21 days) at a dosage of 10 mg/kg. The administration of extract at single dose or repeated doses showed a significant effect in lowering the TC level ($P < 0.05$ and $P < 0.01$) in both normal and STZ-diabetic rats, respectively. However, a significant change in TG was detected only for the repeated dose group ($P < 0.01$), and the extract also prevented weight loss in the diabetic animals [46].

In 2006, Kumari et al. investigated the antihyperlipidemic effect of ethanolic extract of *Eclipta prostrata* in albino rats at different doses (50, 100, 150 and 200 mg/kg) for 30 days. At the doses of 150 and 200 mg/kg, the biochemical parameters of various lipid profiles in serum, liver and heart of the animals showed that the ethanolic extract of *E. prostrata* significantly reduced ($P < 0.001$) the increased levels of total lipids, TC, TG, phospholipids and FFA in a dose-dependent manner [47].

Fruit

Antihyperlipidemic activity of pink guava (*Psidium guajava*) puree was examined in high-fat diet-induced obese rats at 500, 1,000 and 2,000 mg/kg orally via a drinking bottle. High-dosage group (2,000 mg/kg) showed a significant reduction of 34.5 % in TC level ($P < 0.05$) followed by medium-dosage group (23.3 %) and low-dosage group (22.3 %). TG level for all treated groups especially

high-dosage group (43.6 %) showed significant reduction as compared to control. HDL-C level showed an increase in the treated group as compared to control. LDL-C level was significantly decreased in high-dosage group (69.7 %) ($P < 0.05$), medium-dosage group (39.4 %) and low-dosage group (37.1 %) [48].

Aqueous extract of *Terminalia chebula* fruit extract and its combination with Gaumutra were examined for antihyperlipidemic activity in high-cholesterol diet-fed rats for 30 days. Treatment with *T. chebula* (300 mg/kg P.O) and its combination with Gaumutra (30 mg/kg P.O) significantly decreased ($P < 0.05$) serum and tissue TC, LDL-C, VLDL-C, TG and AI, while HDL-C level was increased. Histological study showed that *T. chebula* caused reduced aortic plaque and fatty liver formation, while its combination with Gaumutra showed no significant effect in aorta and liver. The study suggested that *T. chebula* and its combination with Gaumutra were both effective as antihyperlipidemic agents [49].

Ghule et al. evaluated the antihyperlipidemic effect of methanolic extract of *Lagenaria siceraria* fruits (LSFE) at different doses (100, 200 and 300 mg/kg P.O) for 30 days in high-fat diet-induced hyperlipidemic rats [50]. Significant reductions were detected in lipid levels [TC, TG, LDL-C and VLDL-C ($P < 0.001$)] in the LSFE-treated rats when compared to the rats fed with high-fat diet. Conversely, HDL-C level was significantly ($P < 0.001$) increased. Furthermore, LSFE also exhibited significant increase in excretion of bile acids [50].

Methanolic extract of *Randia dumetorum* fruit has been evaluated for its antihyperlipidemic effect on triton WR-1339-induced hyperlipidemic rats at dosage of 400 mg/kg for 2 days. A significant effect on TC level was detected ($P < 0.05$), while no effect was observed on TG level at 18 and 24 h [36]. Ambili et al. evaluated the antihyperlipidemic effect of *Averrhoa bilimbi* L. fruit on Triton X-100-induced hypercholesterolemia and high-fat diet-fed hyperlipidemic rats at dosages of 125 and 250 mg/kg for fruit homogenate and at 25, 50, 100, 200 and 400 mg/kg for aqueous extract. In the acute study, the fruit and its water extract exhibited antihypercholesterolemic activity compared with triton control group. Meanwhile, in chronic high-fat diet-fed hyperlipidemic rats, the fruit (125 mg/kg) as well as its water extract (50 mg/kg) were found to be effective in lowering lipids. It was noted that the optimum antihypercholesterolemic activity was observed at 50 mg/kg. Surprisingly, further increase in the dose does not result in a significant reduction in the cholesterol level [51].

Antihyperlipidemic activity of *Anthocephalus indicus* fruit extract was examined in hyperlipidemic rats fed a triton- and cholesterol-rich high-fat diet. Administration of the fruit extract (500 mg/kg) caused significant decreases in the levels of TC ($P < 0.001$), phospholipids ($P < 0.001$) and TG ($P < 0.001$) [52].

Fruit extract of *Helicteres isora* was examined for its antihyperlipidemic activity in STZ-induced diabetic rats. TC, TG, phospholipids, LDL-C and VLDL-C were increased and the HDL-C level was significantly decreased in diabetic rats. After treated with *H. isora* fruit extract, the lipid levels of diabetic rats were restored to near normal level [53].

Lagenaria siceraria fresh fruits extract was studied for antihyperlipidemic activity using atherogenic diet-induced hyperlipidemic male Wistar rats at dose of 200 mg/kg for 28 days. It was found that *L. siceraria* extract significantly reduced TC, TG and LDL-C and increased HDL-C levels when compared with the hyperlipidemic control groups ($P < 0.05$) [54].

Root

Radix notoginseng (Sanchi) is one of a Chinese traditional medicine prepared from roots of the herb *Panax notoginseng*. It was investigated for its hypolipidemic effect in high-fat diet-induced hyperlipidemic male Sprague–Dawley rats at 0.25, 0.5 and 1 % (w/w) for 4 weeks [55]. A significant decrease in serum levels of TC, TG and LDL-C was observed with an increase in HDL-C level. Also, treated animals showed reduced level of hepatic HMG-Co A reductase. Furthermore, it was found that Sanchi improved hepatic antioxidant status as shown by increased superoxide dismutase and glutathione peroxidase activities and reduced levels of lipid peroxidation. These findings suggest that the consumption of Sanchi could improve lipid profiles, inhibit peroxidation and increase the activity of antioxidant enzymes, which can result in reduction of the risk of coronary heart disease associated with hyperlipidemia and oxidative stress [55].

Chen and Pan were studied the antihyperlipidemic effect of ethanolic extract of *Glycine tomentella* root (GT-E) using tilapia fish as a model. Commercial feed given daily at 2 % body mass for 8 weeks or at 1 % body mass for 12 weeks was used to induce hyperlipidemia and hypercholesterolemia. TC, TG and LDL-C in plasma of the GT-E group were significantly decreased, although plasma total antioxidant status was significantly higher than that of the control group [56]. GT-E showed hypolipidemic and hypocholesterolemic effects and inhibiting LDL oxidation in tilapia similar to the effects in mammals; thus, tilapia was considered as a substitute animal model for prescreening antiatherosclerosis effect of natural products [56].

Seeds

Sesame seeds (*Sesamum indicum*) powder was studied for hypolipidemic activity in male albino rats. *S. indicum* powder was administered at 5 and 10 % dose levels along with

either normal or hypercholesterolemic diet for 4 weeks. Administration of *S. indicum* powder to hypercholesterolemic rats resulted in a significant decline in plasma and hepatic total lipid, TC and LDL-C levels. Furthermore, plasma HDL-C level was increased [57]. Moreover, these animals also had increased fecal excretion of cholesterol, neutral sterol and bile acid along with increases in hepatic HMG-Co A reductase activity and bile acid content. Additionally, sesame seed improved the hepatic antioxidant status (catalase and superoxide dismutase activities) with a reduction in lipid peroxidation. No significant changes in lipid or antioxidant profiles occurred in the normocholesterolemic rats administered with sesame seed powder. The study suggested that these beneficial effects of sesame seed in hypercholesterolemic rats appeared to be due to its fiber, sterols, polyphenols and flavonoids contents enhancing the fecal cholesterol excretion and bile acid production as well as increasing the antioxidant enzyme activities [57].

The antihyperlipidemic effects of the methanolic extract of *Celastrus paniculatus* seeds were examined in high-fat-induced hypercholesterolemic rats. The methanolic extract of *C. paniculatus* seeds at the optimized dose of 65 mg/kg substantially reduced the plasma TC, TG and LDL-C in comparison with hypercholesterolemic control animals group, and the results were comparable with the atorvastatin (5 mg/kg). AI and liver weight of treated animals were significantly reduced, while the HDL-C was substantially increased as compared to control group. Hepatic bile synthesis was enhanced due to a significant increase in the activities of LPL and LCAT [58]. Furthermore, the activities of HMG-Co A reductase, glucose 6-phosphate dehydrogenase and malate dehydrogenase were significantly reduced ($P < 0.05$). Histological studies showed less cholesterol deposits in the aorta of animals fed with seed extract of *C. paniculatus* compared to the induced hypercholesterolemic animals not given *C. paniculatus* supplement [58].

Coriandrum sativum seeds extract was investigated for its antihyperlipidemic properties in cholesterol-fed rabbits. The treatment of 70 % methanolic extract of *C. sativum* seeds at the dose level of 500 mg/kg reduced serum lipid profile and increased HDL-C ratio. *C. Sativum* extract feeding increased the fecal excretion of cholesterol and phospholipids. Histology studies showed less cholesterol deposits in the aorta of high-cholesterol diet animals given *C. sativum* compared to the hyperlipidemic control animals [59].

The aqueous extract of *Aegle marmelos* seeds was assessed for its antihyperlipidemic effect at different doses (100, 250 and 500 mg/kg) in normal as well as sub [fasting blood glucose (FBG) normal; glucose tolerance abnormal] and mild (FBG 120–250 mg/dL) diabetic rats for a period of 14 days. Treatment with a dose of 250 mg/kg reduced the TC level by 25.5 %, increased the HDL-C of 33.4 % and decreased LDL-C and TG of 54.0 and 45.8 %, respectively.

These results confirmed that aqueous seed extract of *A. marmelos* possess hypolipidemic effects in diabetic rats [60].

Kalonji (*Nigella sativa*), an herbal medicine was investigated for its antihyperlipidemic activity in a month study on 60 male and female hyperlipidemic patients of age 22–65 years. The patients were divided in two equal groups, where patients in group I were advised to take two spoons of kalonji, while patients in group II were advised to take placebo capsules after breakfast every day for the period of 4 weeks. Treatment with kalonji significantly reduced LDL-C from 185.2 to 157.7 mg/dL and serum TC from 251.11 to 230.7 mg/dL, and HDL-C level was raised from 31.7 to 36.0 mg/dL [61].

Plant mixture and compounds

Maruthappana and Shree examined the antihyperlipidemic effects of Geriforte, an indigenous herbomineral compound, which is a mixture of several herbal extracts and minerals including amla (*Phyllanthus emblica* Linn.), brahmi (*Centella asiatica* Linn.), asvagandha (*Withania somnifera* Dunal), haldi (*Curcuma longa* Linn.), bhanga (*Eclipta alba* Linn.), lohbbhasma (iron oxide), jasadbbhasma (zinc oxide), onion (*Allium cepa* Linn.), garlic (*Allium sativum* Linn.), grape (*Vitis vinifer* Linn.), carrot (*Docus carota* Linn.) and many other herbs. Hyperlipidemia was induced by oral administration of butter fat diet. Intragastric administration of Geriforte (450 mg/kg) for 8 days significantly decreased body weight, TC, TG, LDL-C and VLDL-C, while significantly increased level of HDL-C. AI was lowered in the Geriforte drug-fed animals when compared to hyperlipidemic animals. Geriforte exhibited competitive potential when compared with the reference herbal formulation Ayurslim [62].

The hypolipidemic effect of a Fatclean formula was examined in a high-fat diet-induced hyperlipidemic rats at 5 % w/w supplement for 6 weeks. Fatclean formula significantly lowered final body weights and visceral fat-pads weights, TC and TG levels, hepatic TC and TG levels, and HMG-Co A and acyl-coenzyme A: cholesterol acyltransferase (ACAT) activities. Furthermore, LPL activity was also significantly increased in the Fatclean group, and the ratios of HDL-C/TC and AI were significantly improved ($P < 0.05$) in the Fatclean group compared with the high-fat diet control group [63].

In a randomized, placebo-controlled double-blind monocentric study, 50 subjects were assigned to 200 mL of soy drink either enriched with 2.6 g plant sterol esters (1.6 g free plant sterol equivalents) or without plant sterols (control) for 8 weeks. Treatment with the plant sterol-enriched soy drink significantly reduced LDL-C by 0.29 mmol/L or 7 % compared with baseline ($P < 0.05$). TC and non-HDL cholesterol concentrations decreased by 0.26 and 0.31 mmol/L ($P < 0.05$), respectively [64].

Essential oil from *O. sanctum* leaves was evaluated for the antihyperlipidemic ability in rats fed with a high-cholesterol diet. The essential oil significantly decreased serum TC, LDL-C, TG and AI, whereas had no significant effect on HDL-C. The essential oil depressed the high levels of liver TC and TG, but had no significant effect on excretion of both lipids in feces. The major constituents of the essential oil of *O. sanctum* leaves were found to be phenylpropanoid compounds (eugenol and methyl eugenol) [11]. The essential oil exhibited a better lipid-lowering activity in comparison with the leaves methanol extract. Apricot kernel and pumpkin kernel oils were assessed for their lipid-lowering activity in rats fed with high-fat diets over a 28 days period. The rats treated with apricot kernel and pumpkin kernel oils showed significantly lowered levels of TC, TG and LDL-C, while the level of HDL-C was raised [65].

Allicin [(R, S)-diallyldisulfid-S-oxide], a sulfur containing compound from garlic, is reported to be an active in atherosclerosis. Allicin at doses of 5, 10, or 20 mg/kg was given orally daily for 12 weeks to male ICR mice. Following a 12-week experimental period, the body weights of allicin fed mice were less than those of control mice on a high-cholesterol diet by 38.24 % ($P < 0.001$) with 5 mg/kg allicin, 39.28 % ($P < 0.001$) with 10 mg/kg allicin and 41.18 % ($P < 0.001$) with 20 mg/kg allicin, respectively [66].

Qi et al. evaluated the antihyperlipidemic activity of natural polysaccharide (ulvan) and its high-sulfate content derivative (HU) in high-cholesterol diet-fed mice for 21 days. They noticed the obvious differences in antihyperlipidemic activity between natural ulvan and HU, strongest antihyperlipidemic activity was found out with HU at the dose of 250 mg/kg compared with natural ulvan fed group, TG and LDL-C concentrations were found to be significantly decreased 28.1 % ($P < 0.05$) and 28.4 % ($P < 0.01$), respectively. But the antihyperlipidemic activity was not seen in dose-dependent way. The authors attributed the antihyperlipidemic activity might be due to high-sulfate content of the polysaccharide [67]. Liu et al. evaluated the polysaccharides from *Apostichopus japonicus* for their in vitro antioxidant and antihyperlipidemic activities in high-fat diet-induced hyperlipidemia on male Wistar rats for 28 days. The antihyperlipidemic effect of polysaccharides was found to be strongest at the dose of 400 mg/kg where TC, TG and LDL-C concentrations decreased significantly by 17.23, 20.78 and 31.18 %. On the other hand, lower dose found to be less effective and higher dose was found to have optimal effect on TC and LDL-C only. The polysaccharides also showed potent antioxidant activity and reducing power [68].

Red ginseng's acidic polysaccharide was administered orally (100–1,000 mg/kg) to triton WR-1339- or corn oil-induced hyperlipidemic rats. In triton-induced animals, red ginseng's acidic polysaccharide significantly reduced

the levels of non-esterified fatty acids and TG in a dose-dependent manner. TC was not reduced while phospholipids levels increased under the same condition. Reductions were also observed in the levels of total lipids. The exogenous hyperlipidemic rat condition triggered by corn oil also supported the antihyperlipidemic activity of red ginseng acidic polysaccharide in serum and hepatic parameters of TG and non-esterified fatty acids. Red ginseng's acidic polysaccharide significantly enhanced the serum activity of LPL, in a dose-dependent manner up to 80 %. Thus, the study suggests that red ginseng's acidic polysaccharide is a potential antihyperlipidemic agent [69]. Ban et al. investigated the effect of dietary feeding of hydroxyethyl methyl cellulose (HEMC) with different viscosities on the body weight and lipid metabolism in high-fat-fed mice for 6 weeks. It was noticed that the addition of HEMC in the diet counteracted this high-fat-induced hyperlipidemia via inhibition of lipogenesis and regulation of adipokine production. It was concluded that the HEMC with high viscosity may be effective in lowering the risk of hyperlipidemia and obesity on high-fat diet model [70].

The effect of water soluble polyphenolic extract of peanut skin (PE) of *Arachis hypogaea* was investigated for its hypolipidemic properties in high-fat diet-induced hyperlipidemic rats (Western diet) 150 and 300 mg/kg for 7 weeks. Significant reductions were observed in the body weight ($P < 0.05$), epididymal fat, TC and TG in liver and plasma ($P < 0.01$), while fecal secretion of TC and TG were significantly ($P < 0.05$) increased after administration of the PE in the group which received 300 mg/kg [71]. The expression of liver mRNA enzymes involved in fatty acid synthesis, such as fatty acid synthase, sterol receptor element-binding protein-1c, acetyl-CoA carboxylase and lipid uptake genes, such as PPAR γ , was decreased, while PPAR α was up-regulated by administration of PE. It has been concluded that the administration of PE can be helpful in improving the lipid homeostasis in rats [71].

Mathew et al. investigated the effect of leucodelphinidin derivative isolated from the bark of *Ficus bengalensis* and quercetin in hypercholesterolemic rats at a dose of 100 mg/kg. Significant reductions in serum TC, LDL-C and increased HDL-C level were noticed in both flavonoid-treated groups when compared to the cholesterol diet-fed rats ($P < 0.01$). In the rats treated with both flavonoids, the AI was significantly reduced comparing with the control group ($P < 0.01$). The concentration of total bile acids in the liver and the fecal excretion of bile acids and neutral sterols was found to be increased in the rats fed with cholesterol containing diet as compared to those fed normal diet [72]. Administration of flavonoids increased the fecal excretion of bile acids and neutral sterols and significantly increased ($P < 0.01$) the concentrations of hepatic bile

acids, as compared to the control cholesterol diet-fed group (leucodelphinidin derivative showed a slightly better effect than quercetin).

The flavonoids were unable to inhibit the activity of 3-hydroxy-3-methyl glutaryl coenzyme A (HMG-Co A) reductase. However, increased cholesterogenesis was observed on treatment with leucodelphinidin and quercetin. Thus, the study suggested that the flavonoids exerted their hypocholesterolemic effects by increasing fecal bile acids and cholesterol excretion [72].

Total flavonoids extracted from corn silk by microwave-assisted extraction using ethanol were tested for their anti-hyperlipidemic effect in high-fat diet-induced hyperlipidemic animals. The total flavonoids significantly ($P < 0.05$) reduced the TC, TG and LDL-C levels [73]. Chen and Li examined the hypolipidemic effect of total flavonoids from mulberry (*Morus alba* L.) leaves in triton WR-1339-induced hyperlipidemic mice. The levels of TG, TC and LDL-C were significantly reduced from 540, 464 and 299 mg/dL to 388, 257 and 189 mg/dL, respectively. It was noticed that the ratios of HDL-C/TC and HDL-C/LDL cholesterol were increased from 33 and 52 % to 42 and 57 %, respectively, indicating additional beneficial effects [74].

Narender et al. identified an alkaloidal amide, aegeline from the plant *A. marmelos* leaves and examined its anti-hyperlipidemic and antihyperglycemic in triton-induced hyperlipidemia model for 18 h using a dose of 100 mg/kg P.O. In addition, the article describes screening for in vivo antihyperlipidemic activity of synthesized series of alkaloidal amides related to aegeline [*N*-(2-hydroxy-2-*p*-tolylethyl)-amides and *N*-(2-oxo-2-*p*-tolylethyl)-amide derivatives]. Three synthetic compounds showed potent activity compared to the natural product aegeline. These compounds also showed strong antioxidant activity, which support their antihyperlipidemic activity. One of the synthetic compounds showed better antihyperlipidemic and antioxidant activities in comparison with the natural products [75].

Karthikesan et al. tested the efficacy of chlorogenic acid and tetrahydrocurcumin alone and in combination in streptozotocin–nicotinamide-induced type 2 diabetic rats. Combined administration of chlorogenic acid (5 mg/kg) and tetrahydrocurcumin (80 mg/kg) for 45 days significantly reduced the STZ-induced changes in lipids, lipoproteins and lipid-metabolizing enzymes when compared to the effects of chlorogenic acid or tetrahydrocurcumin alone in diabetic rats [76]. Antihyperlipidemic effect of gallic acid (GA) isolated from fruit rind of *Terminalia bellerica* was assessed at the doses of 5, 10 and 20 mg/kg P.O for 28 days in STZ-treated diabetic male Wistar rats, and it was compared with the effect of synthetic gallic acid procured from Sigma-Aldrich (USA) [77]. GA at a dose of 20 mg/kg showed significant decreased of TC, TG, LDL-C, urea, uric

acid, creatinine and simultaneously markedly increased plasma insulin, C-peptide and glucose tolerance levels. Also, GA restored the total protein, albumin and body weight of diabetic rats to near normal. Thus, the results indicate that gallic acid present in fruit rind of *T. bellerica* is responsible for the regeneration of β -cells and normalizing all the biochemical parameters related to the patho-biochemistry of diabetes mellitus, and hence, it could be used as a potent antidiabetic agent [77].

Protocatechuic acid, an important constituent of *H. sabdariffa* flowers at doses of 25 and 50 mg/kg was evaluated in fructose-induced hyperlipidemic rats. It showed a significant decrease in a dose-dependent manner in the levels of serum TC, TG, LDL-C and glucose ($P < 0.05$, $P < 0.01$, $P < 0.05$ and $P < 0.01$, respectively). In addition, HDL-C level was significantly increased ($P < 0.01$) in serum as compared to hyperlipidemic control group [78].

In a pilot study, lipid-lowering activity of berberine was evaluated in obese human subjects (Caucasian) and rats. Human subjects were given 500 mg berberine orally three times a day for 12 weeks, while the rats were administered 380 mg/kg of berberine for 2 weeks. The results demonstrated that berberine treatment produced a mild weight loss (average 5 lb subject) in obese human subjects. Interestingly, the treatment significantly reduced blood lipid levels (23.0 % decrease in TG and 12.2 % decrease in cholesterol levels) in human subjects. Meanwhile, 34.7 % reduction in TG and 9 % in TC levels were observed in rats [79].

Conclusion

From the total number of investigations, it is found that triton WR-1339 (28 %)- and high-fat diet-induced hyperlipidemic models (44 %) are the popularly employed models for the evaluation of antihyperlipidemic potencies of plant extracts, compounds or formulations thereof, and the most commonly used laboratory animals for these purposes were either Wistar or albino rats (preferably male). The study duration selected for the evaluation of the candidate natural products was 48 h for triton-induced model and for high-fat diet model it was 4–6 weeks. The most commonly evaluated doses of the natural products ranged from 200 to 800 mg/kg, but it is noteworthy to mention that the dose of natural product under investigation is basically dependent on the nature of secondary metabolite(s) present and hence cannot be generalized for the other studies. In few cases, we found that lower doses as low as 50–200 mg/kg were also effective in reducing the lipid levels; hence, the dose and therapeutic efficacy of the natural products are solely attributed by the secondary metabolite(s) present in the plants under investigation.

From the literatures reviewed, it seems that the production of secondary metabolites responsible for the antihyperlipidemic activity is more in leaves and aerial parts of the medicinal plants, which are metabolically active sites compared with other parts of plants. But we could neither find clear description about this fact in any of the literature nor strong correlation from the publications to support this hypothesis. We also faced ambiguity while analyzing the data with pure compounds and fractions from the plant, because of less number of articles. The occurrence of secondary metabolites responsible for lipid-lowering activity might seem to be closely associated within the plant families, namely *Amaranthus*, *Lamiaceae*, *Asteraceae*, *Malvaceae*, *Myrtaceae*, *Fabaceae* and *Apiaceae*. Further chemotaxonomical research within these families will be of great advantage for future research toward the development of novel antihyperlipidemic agents. The search for novel lipid-lowering agents from plant sources is going on with ray of hope of getting better candidate with multiple advantages not only as lipid-lowering agent and but also for reducing the several risks factors associated with cardiovascular disorders originating from hyperlipidemia. Developments happening in this area of research might bring hopefully promising results in the coming decades.

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Conflict of interest None.

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